

# Design Rules Summary

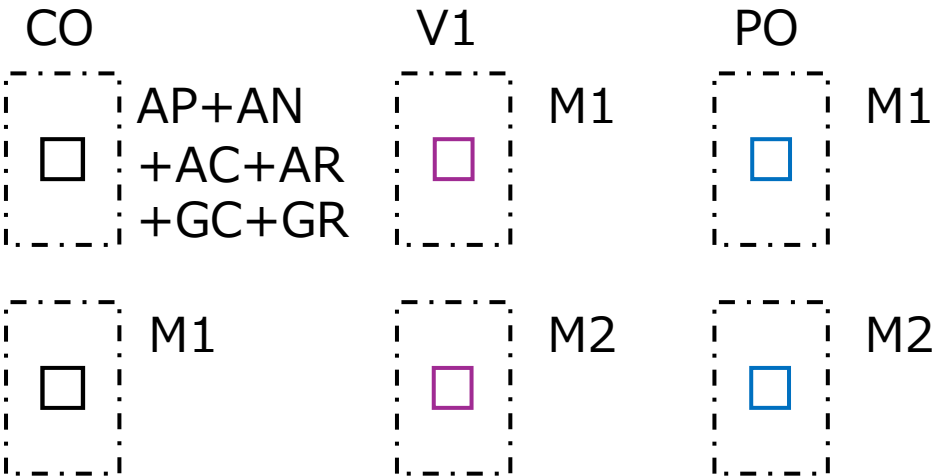
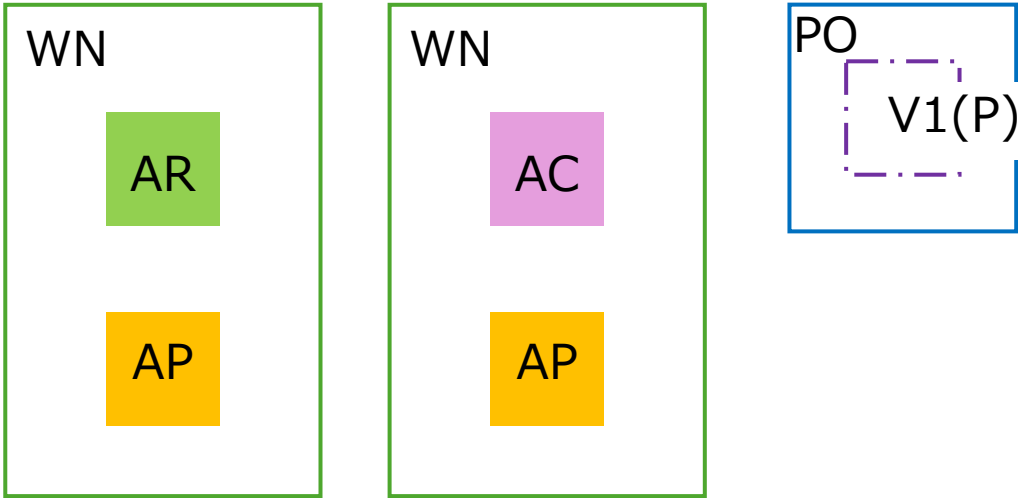
for Drawing Layers

2026/01/29

# Drawing Layer Table

|    | Original MASK layers |     |      |  | Proposed Drawing layers |     |      |                           |  |  |
|----|----------------------|-----|------|--|-------------------------|-----|------|---------------------------|--|--|
|    | Name                 | NUM | TYPE |  | Name                    | NUM | TYPE |                           |  |  |
| 1  | PSUB                 | 140 | 0    |  | WN                      | 140 | 0    |                           |  |  |
| 2  | NW                   | 36  | 0    |  |                         |     |      | MDP generatioed           |  |  |
| 3  | HVNW                 | 141 | 0    |  |                         |     |      | MDP generatioed           |  |  |
| 4  | L                    | 3   | 0    |  |                         |     |      | MDP generatioed           |  |  |
|    |                      |     |      |  | AP                      | 3   | 1    | Active for P+             |  |  |
|    |                      |     |      |  | AN                      | 3   | 2    | Active for N+             |  |  |
|    |                      |     |      |  | AR                      | 3   | 3    | Active for resistor       |  |  |
|    |                      |     |      |  | AC                      | 3   | 4    | Active for capacitance    |  |  |
| 5  | NF                   | 25  | 0    |  |                         |     |      | MDP generatioed           |  |  |
| 6  | PF                   | 26  | 0    |  |                         |     |      | MDP generatioed           |  |  |
| 7  | CL                   | 143 | 0    |  |                         |     |      | MDP generatioed           |  |  |
| 8  | HPBE                 | 144 | 0    |  |                         |     |      | No USE                    |  |  |
| 9  | HNBE                 | 145 | 0    |  |                         |     |      | No USE                    |  |  |
| 10 | PBE                  | 146 | 0    |  |                         |     |      | MDP generatioed           |  |  |
| 11 | NBE                  | 147 | 0    |  |                         |     |      | MDP generatioed           |  |  |
| 12 | HPM                  | 33  | 0    |  |                         |     |      | MDP generatioed           |  |  |
| 13 | RHP                  | 148 | 0    |  |                         |     |      | No USE                    |  |  |
| 14 | SG                   | 8   | 0    |  |                         |     |      |                           |  |  |
|    |                      |     |      |  | GC                      | 8   | 1    | Gate Poly transistor      |  |  |
|    |                      |     |      |  | GR                      | 8   | 2    | Gate POLY resistor        |  |  |
| 15 | PM                   | 35  | 0    |  |                         |     |      | MDP generatioed           |  |  |
| 16 | NM                   | 7   | 0    |  |                         |     |      | MDP generatioed           |  |  |
| 17 | R                    | 12  | 0    |  |                         |     |      | MDP generatioed           |  |  |
| 18 | PSD                  | 9   | 0    |  |                         |     |      | MDP generatioed           |  |  |
| 19 | NSD                  | 28  | 0    |  |                         |     |      | MDP generatioed           |  |  |
| 20 | CONT                 | 11  | 0    |  | CO                      | 11  | 0    |                           |  |  |
| 21 | M1                   | 13  | 0    |  | M1                      | 13  | 0    |                           |  |  |
| 22 | TC                   | 19  | 0    |  | V1                      | 19  | 0    |                           |  |  |
| 23 | M2                   | 20  | 0    |  | M2                      | 20  | 0    |                           |  |  |
| 24 | PRO                  | 14  | 0    |  | PO                      | 14  | 0    |                           |  |  |
|    |                      |     |      |  |                         |     |      |                           |  |  |
|    | TXM1                 | 48  | 0    |  | TXM1                    | 48  | 0    | M1 Label                  |  |  |
|    | PIN                  | 48  | 1    |  | PIN                     | 48  | 1    | M1 Pin                    |  |  |
|    | TXM2                 | 49  | 0    |  | TXM2                    | 49  | 0    | M2 Label                  |  |  |
|    | MASK                 | 63  | 0    |  | MASK                    | 63  | 0    | Waiver region             |  |  |
|    | PTECT                | 63  | 1    |  | PTECT                   | 63  | 1    | Protect region (LOGO)     |  |  |
|    | SCRB                 | 80  | 0    |  | SCRB                    | 80  | 0    | Waiver region (EDGE SEAL) |  |  |
|    | prBoundary           | 235 | 0    |  | prBoundary              | 235 | 0    | Cell boundary             |  |  |

# Prohibited



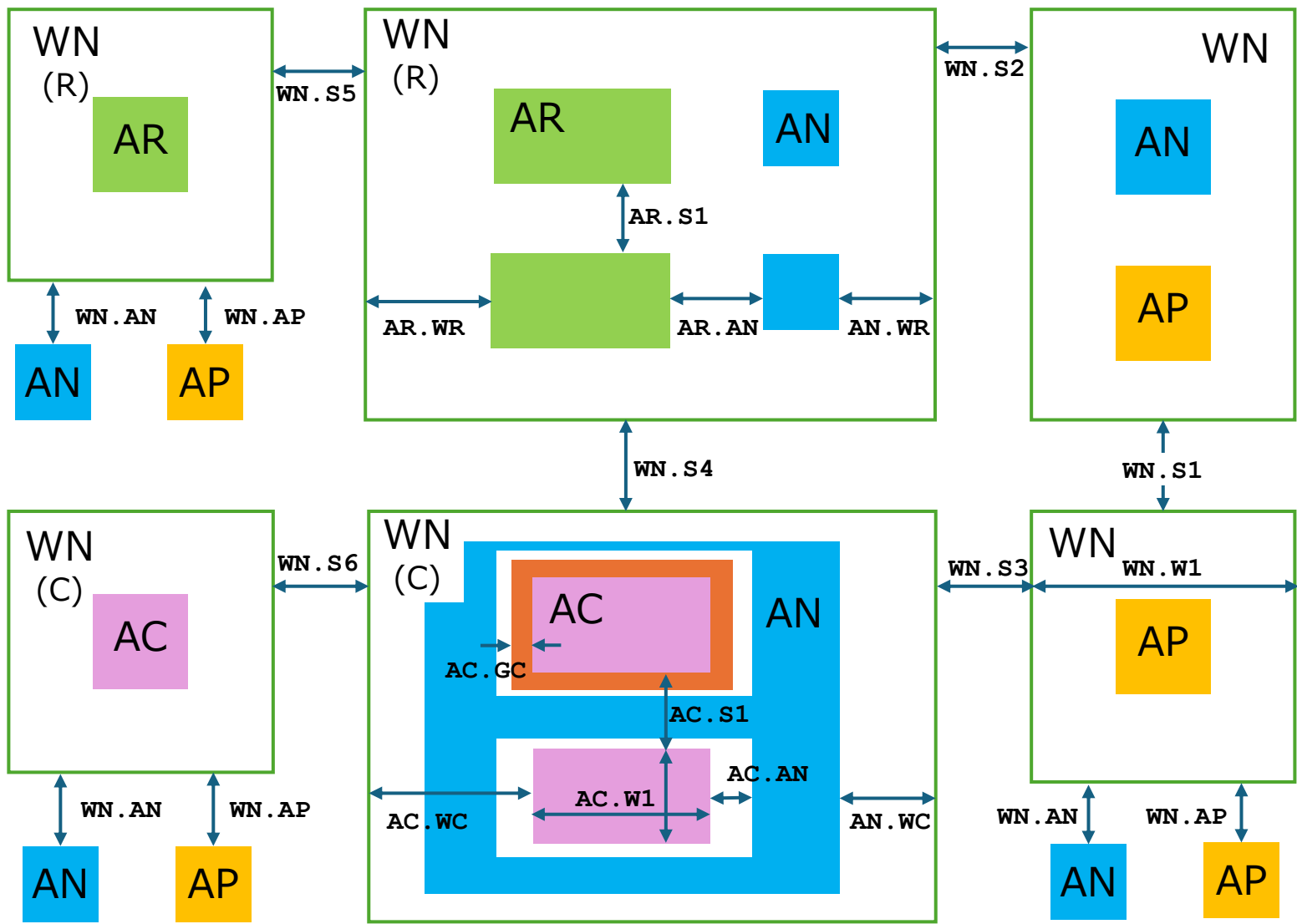
| Rule  | L1     | L2       | Func  | MIN | MAX |
|-------|--------|----------|-------|-----|-----|
| WN.Z1 | WN (R) | AP+AC    | None  | 0.0 |     |
| WN.Z2 | WN (C) | AP+AR    | None  | 0.0 |     |
| CO.Z1 | CO     | AA+GC+GR | None  | 0.0 |     |
| CO.Z3 | CO     | M1       | None  | 0.0 |     |
| V1.Z1 | V1     | M1       | None  | 0.0 |     |
| V1.Z2 | V1     | M2       | None  | 0.0 |     |
| PO.Z1 | PO     | M1       | None  | 0.0 |     |
| PO.Z2 | PO     | M2       | None  | 0.0 |     |
| PO.Z3 | PO     | V1 (P)   | Exist | 0.0 |     |

**Function:**

**Wmin/max:** L1 Width min/max  
**Smin/max:** L1 Space min/max (L1=L2)  
**Smin/max:** L1 Separate L2 min/max  
**Emin/max:** L1 Enclosed L2 min/max  
**Fmin/max:** L1 Enclosing L2 min/max  
**ECmin:** MOS Endcap min  
**Wfix:** L1 Windth equal  
**Sfix:** Space equal  
**Donut:** L1 Souuounded by L2  
**None:** Not Exist

V1 (P) : V1 on PE

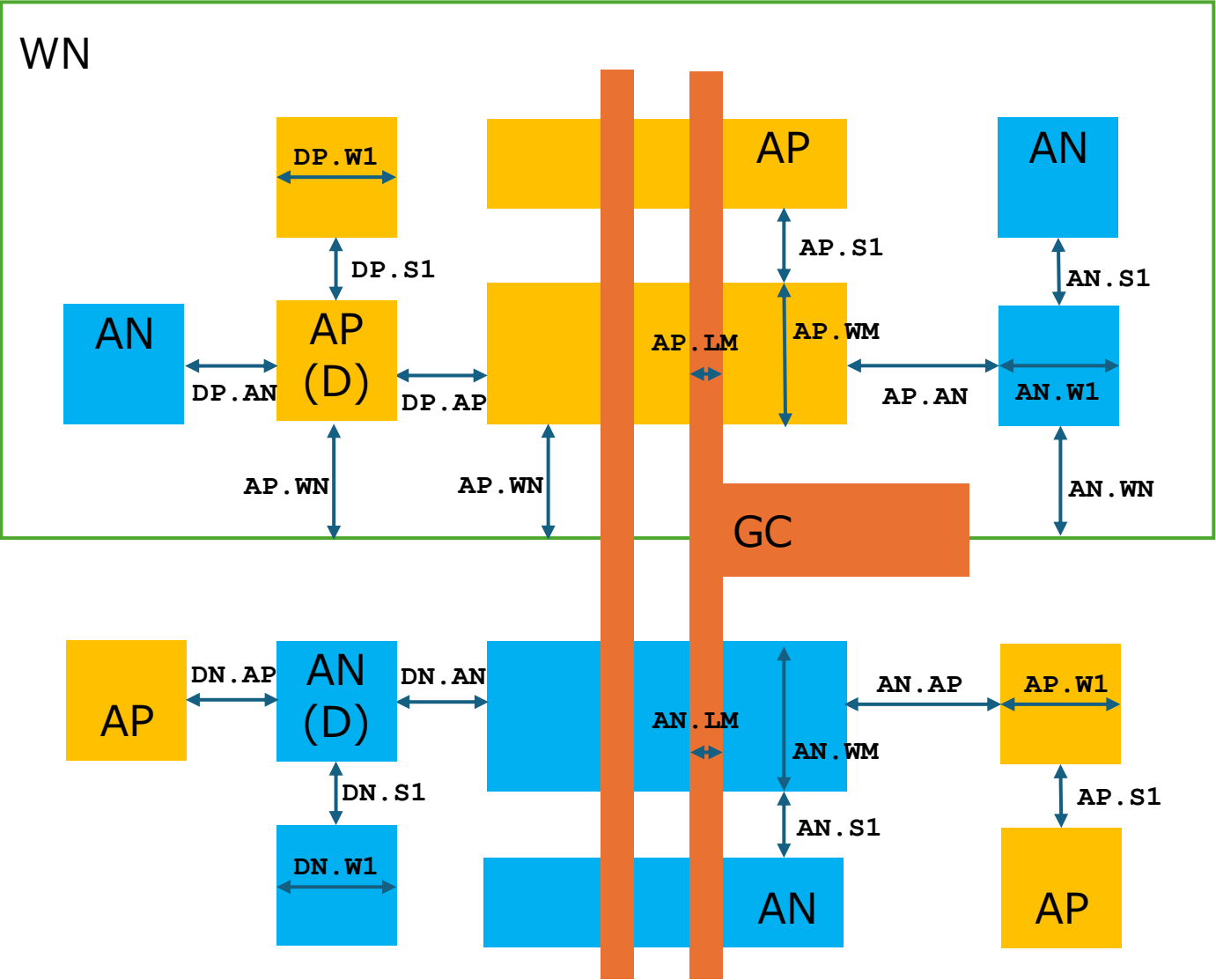
# WN to AP/AN/AR/AC related



| Rule  | L1     | L2     | Func     | MIN  | MAX   |
|-------|--------|--------|----------|------|-------|
| WN.W1 | WN     |        | Wmin     | 8.0  |       |
| WN.S1 | WN     | WN     | Smin     | 12.0 |       |
| WN.S2 | WN     | WN (R) | Smin     | 9.5  |       |
| WN.S3 | WN     | WN (C) | Smin     | 12.0 |       |
| WN.S4 | WN (R) | WN (C) | Smin     | 9.5  |       |
| WN.S5 | WN (R) | WN (R) | Smin     | 8.0  |       |
| WN.S6 | WN (C) | WN (C) | Smin     | 12.0 |       |
| WN.AP | WN     | AP     | Smin     | 5.0  |       |
| WN.AN | WN     | AN     | Smin     | 10.0 |       |
| AR.S1 | AR     | AR     | Smin     | 4.0  |       |
| AR.WR | AR     | WN (R) | Emin     | 10.0 |       |
| AN.WR | AN     | WN (R) | Emin     | 5.0  |       |
| AR.AN | AR     | AN     | Smin     | 4.0  |       |
| AC.W1 | AC     |        | Wmin/max | 28.5 | 120.0 |
| AC.S1 | AC     | AC     | Smin     | ???  | ???   |
| AC.WC | AC     | WN (C) | Emin     | ???  | ???   |
| AN.WC | AN     | WN (C) | Emin     | 3.0  |       |
| AC.AN | AC     | AN     | Smin     | 2.8  |       |
| AC.GC | AC     | GC     | Emin     | 1.4  |       |
| AC.R1 | AC     | AN (C) | Donut    | 0.0  |       |

WN(R) : WN overlapped AR  
WN(C) : WN overlapped AC  
AN(C) : AN in WN(C)

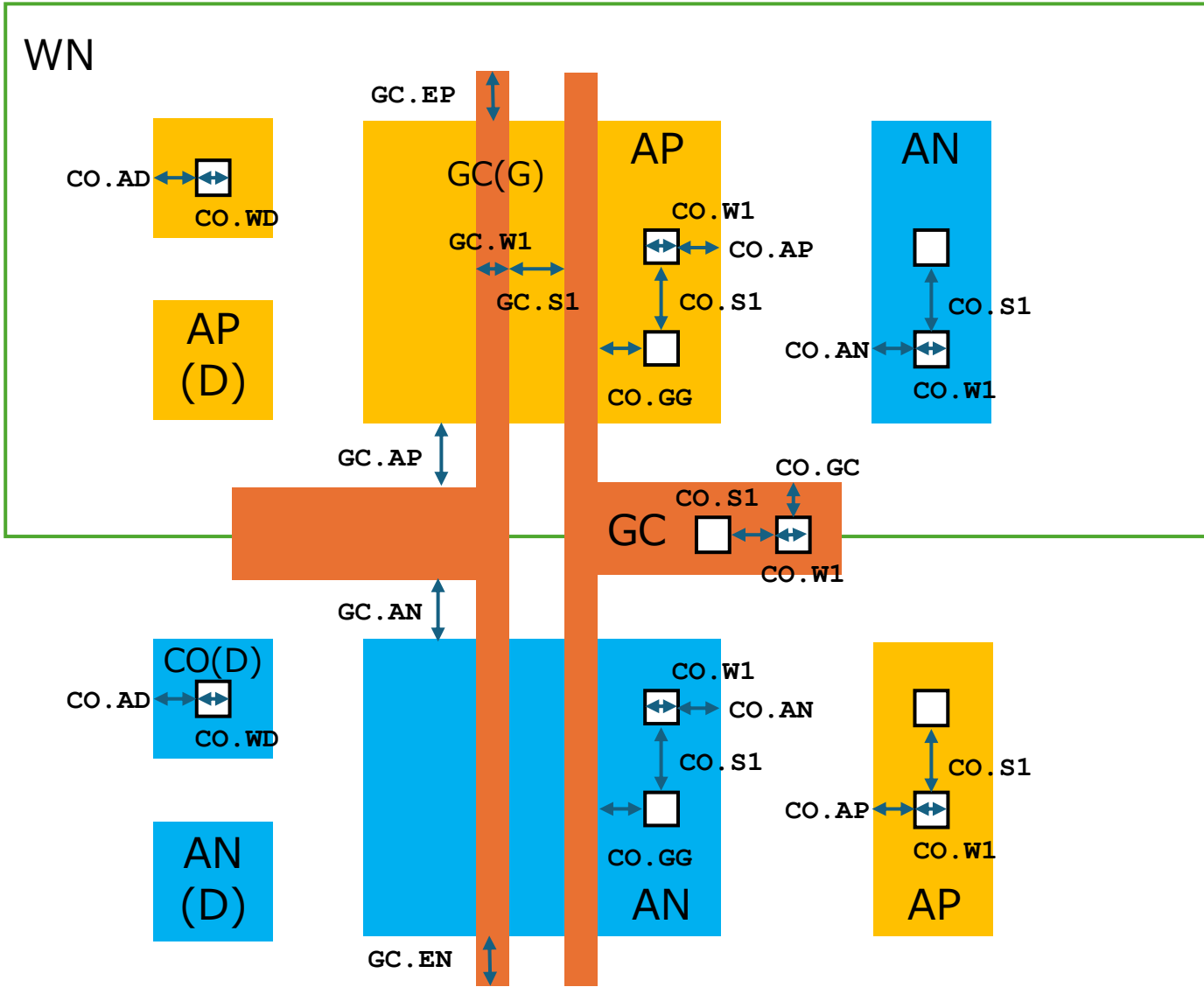
# MOS transistor related



| Rule  | L1   | L2     | Func     | MIN | MAX  |
|-------|------|--------|----------|-----|------|
| AP.W1 | AP   |        | Wmin     | 1.4 |      |
| AP.S1 | AP   | AP     | Smin     | 1.4 |      |
| AP.AN | AP   | AN     | Smin     | 2.8 |      |
| AP.WM | PMOS | AP-GC  | Wmin/max | 3.4 | 60.0 |
| AP.LM | PMOS | GC-AN  | Lmin/max | 1.0 | 30.0 |
| AP.WN | AP   | WN     | Emin     | 7.0 |      |
| DP.W1 | DP   |        | Wmin     | 2.8 |      |
| DP.S1 | DP   | DP     | Smin     | ??? | ???  |
| DP.AP | DP   | AP     | Smin     | 2.8 |      |
| DP.AN | DP   | AN     | Smin     | 2.8 |      |
| AN.W1 | AN   |        | Wmin     | 1.4 |      |
| AN.S1 | AN   | AN     | Smin     | 1.4 |      |
| AN.AP | AN   | AP     | Smin     | 2.8 |      |
| AN.WM | NMOS | AN-GC  | Wmin/max | 3.4 | 60.0 |
| AN.LM | NMOS | GC-AN  | Lmin/max | 1.0 | 30.0 |
| AN.WN | AN   | WN (M) | Emin     | 5.0 |      |
| DN.W1 | DN   |        | Wmin     | 2.8 |      |
| DN.S1 | DN   | DN     | Smin     | ??? | ???  |
| DN.AP | DN   | AP     | Smin     | 2.8 |      |
| DN.AN | DN   | AN     | Smin     | 2.8 |      |

WN(M) : WN overlapped AN/AP

# Contact and Poly related



| Rule  | L1     | L2     | Func  | MIN | MAX  |
|-------|--------|--------|-------|-----|------|
| GC.W1 | GC     |        | Wmin  | 1.0 | 30.0 |
| GC.S1 | GC     | GC     | Smin  | 1.2 |      |
| GC.AP | GC     | AP     | Smin  | 0.4 |      |
| GC.AN | GC     | AN     | Smin  | 0.4 |      |
| GC.EP | PMOS   | GC     | ECmin | 1.2 |      |
| GC.EN | NMOS   | GC     | ECmin | 1.2 |      |
| CO.W1 | CO     |        | Wmin  | 1.0 |      |
| CO.S1 | CO     | CO     | Smin  | 1.0 |      |
| CO.WD | CO (D) |        | Wfix  | 1.2 |      |
| CO.AD | CO (D) | AP+AN  | Emin  | 1.2 |      |
| CO.AP | CO     | AP     | Emin  | 0.8 |      |
| CO.AN | CO     | AN     | Emin  | 0.8 |      |
| CO.GG | CO     | GC (G) | Smin  | 1.0 |      |
| CO.GC | CO     | GC+GR  | Emin  | 0.8 |      |

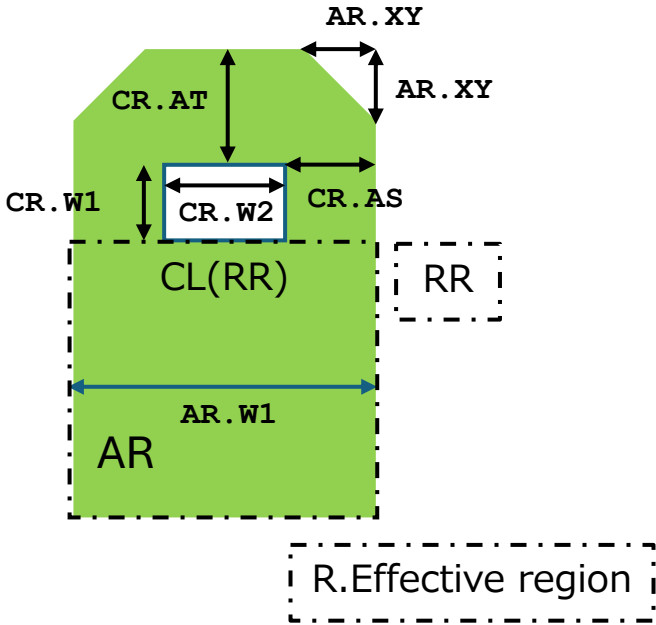
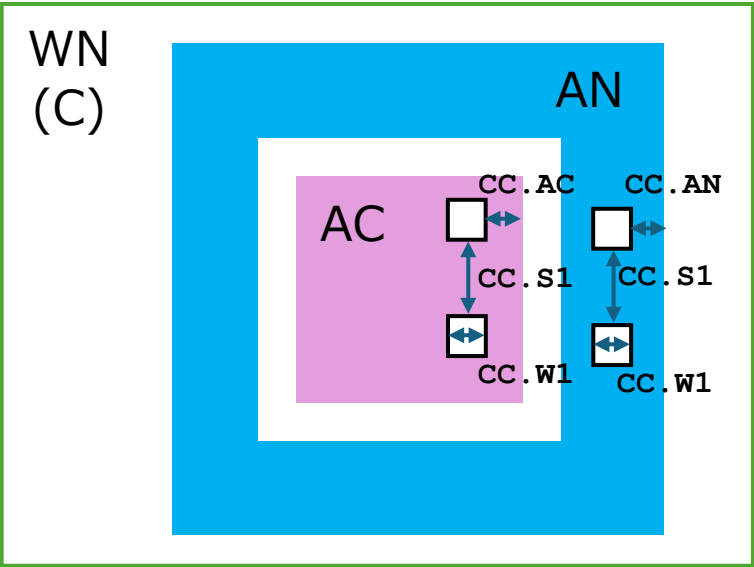
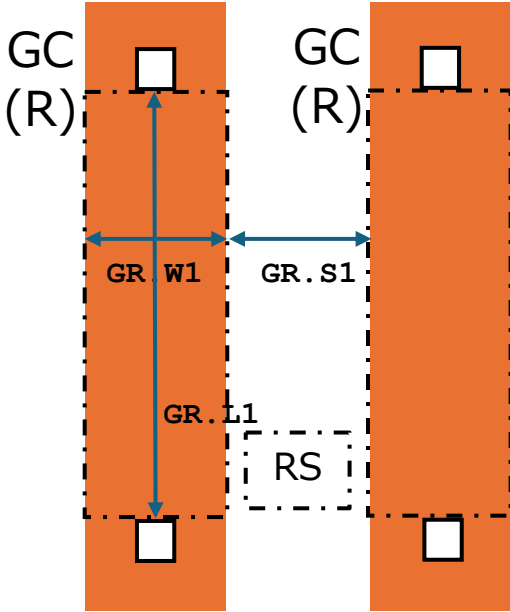
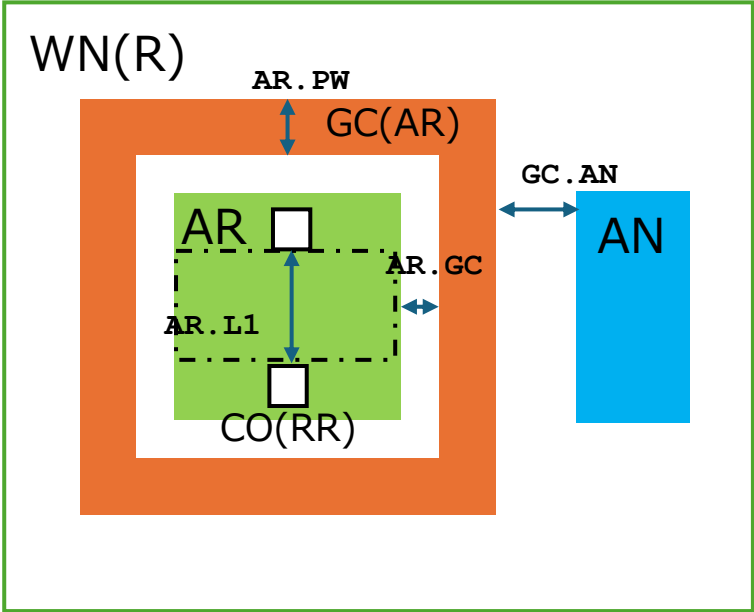
AP (D) : AP not interacted GC

AN (D) : AN not interacted GC

$$\text{CO}(D) : \text{CO on AP}(D) / \text{AN}(D)$$

GC (G) : GC for PMOS/NMOS

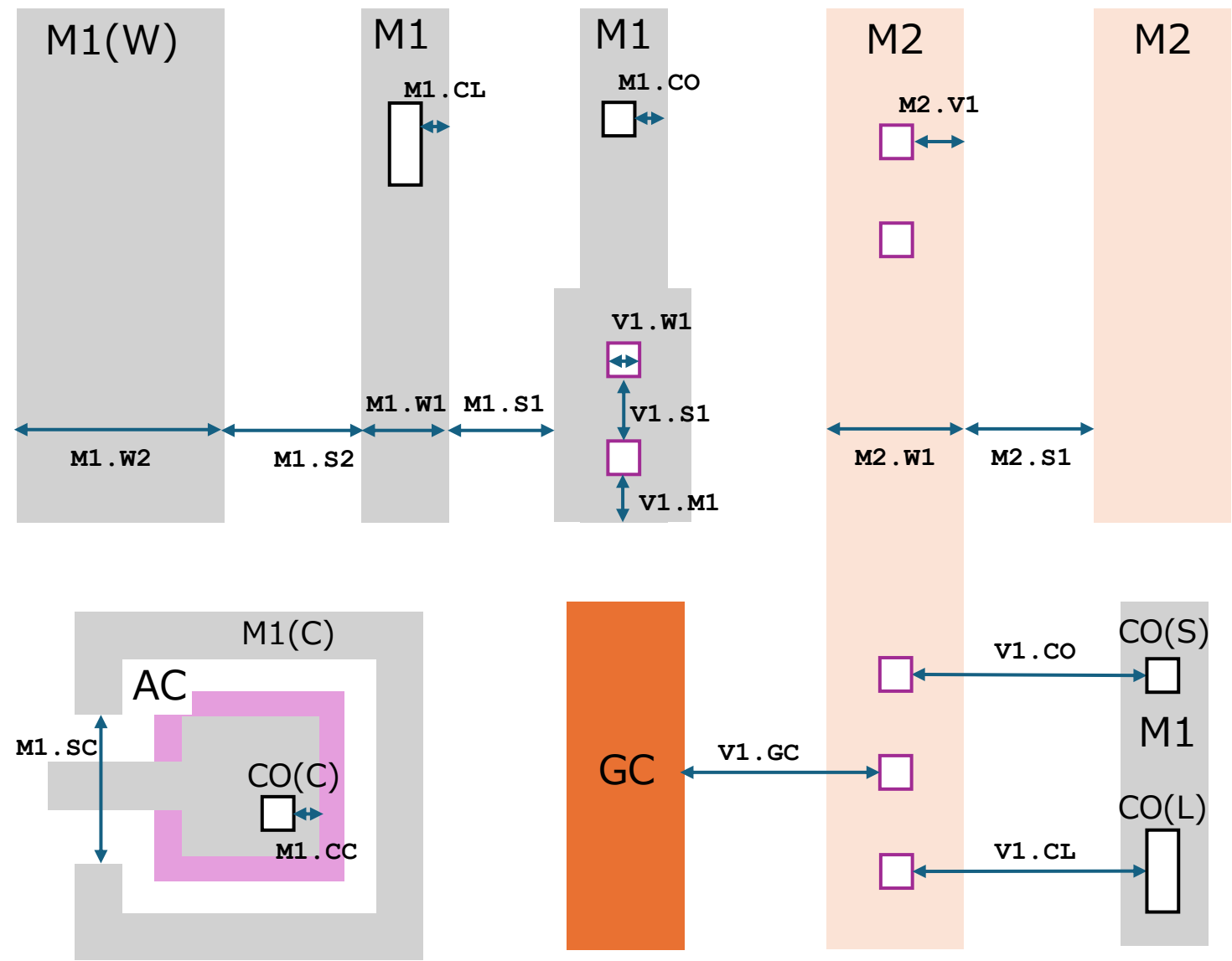
# Resistor(AR/RS) and Capacitor(AC) related



| Rule  | L1      | L2      | Func     | MIN  | MAX   |
|-------|---------|---------|----------|------|-------|
| AR.W1 | RR      | RR (L)  | Wmin/max | 2.8  | 20.0  |
| AR.L1 | RR      | RR (W)  | Lmin/max | 13.0 | 100.0 |
| AR.GC | GC      | AR      | Smin     | 1.0  |       |
| AR.PW | GC (AR) |         | Wmin     | 2.0  |       |
| AR.XY | AR      | Bevel   | XYmin    | 1.0  |       |
| GC.AN | GC (AR) | AN      | Smin     | ???  | ???   |
| GC.R1 | AR      | GC (AR) | Donut    | 0.0  |       |
| GC.R2 | GC      | TieDown |          | 0.0  |       |
| GR.W1 | RS      | RS (L)  | Wmin/max | 4.0  | 20.0  |
| GR.L1 | RS      | RS (W)  | Lmin/max | 20.0 | 100.0 |
| GR.S1 | RS      | RS      | Smin     | 2.0  |       |
| CR.W1 | CO (RR) |         | Wmin     | 1.0  |       |
| CR.W2 | CL (RR) |         | Wmin/max | 1.0  | 97.5  |
| CR.AT | CO (RR) | AR (T)  | Emin     | 1.5  |       |
| CR.AS | CO (RR) | AR (S)  | Emin     | 1.0  |       |
| CC.W1 | CO (C)  |         | Wmin     | 1.2  |       |
| CC.S1 | CO (C)  | CO (C)  | Smin     | 1.2  |       |
| CC.AC | CO (C)  | AC      | Emin     | 2.9  |       |
| CC.AN | CO (C)  | AN      | Emin     | 1.2  |       |

GC (AR) : GC in WN(R)  
RS(L) : Length of AR effective  
RS(W) : Width of AR effective  
CO(RR) : CO short side on AR  
CL(RR) : CO large side on AR  
AR(T) : AR top fringe  
AR(S) : AR side fringe  
CO(C) : CO on AC

# M1 /V1 /M2 related

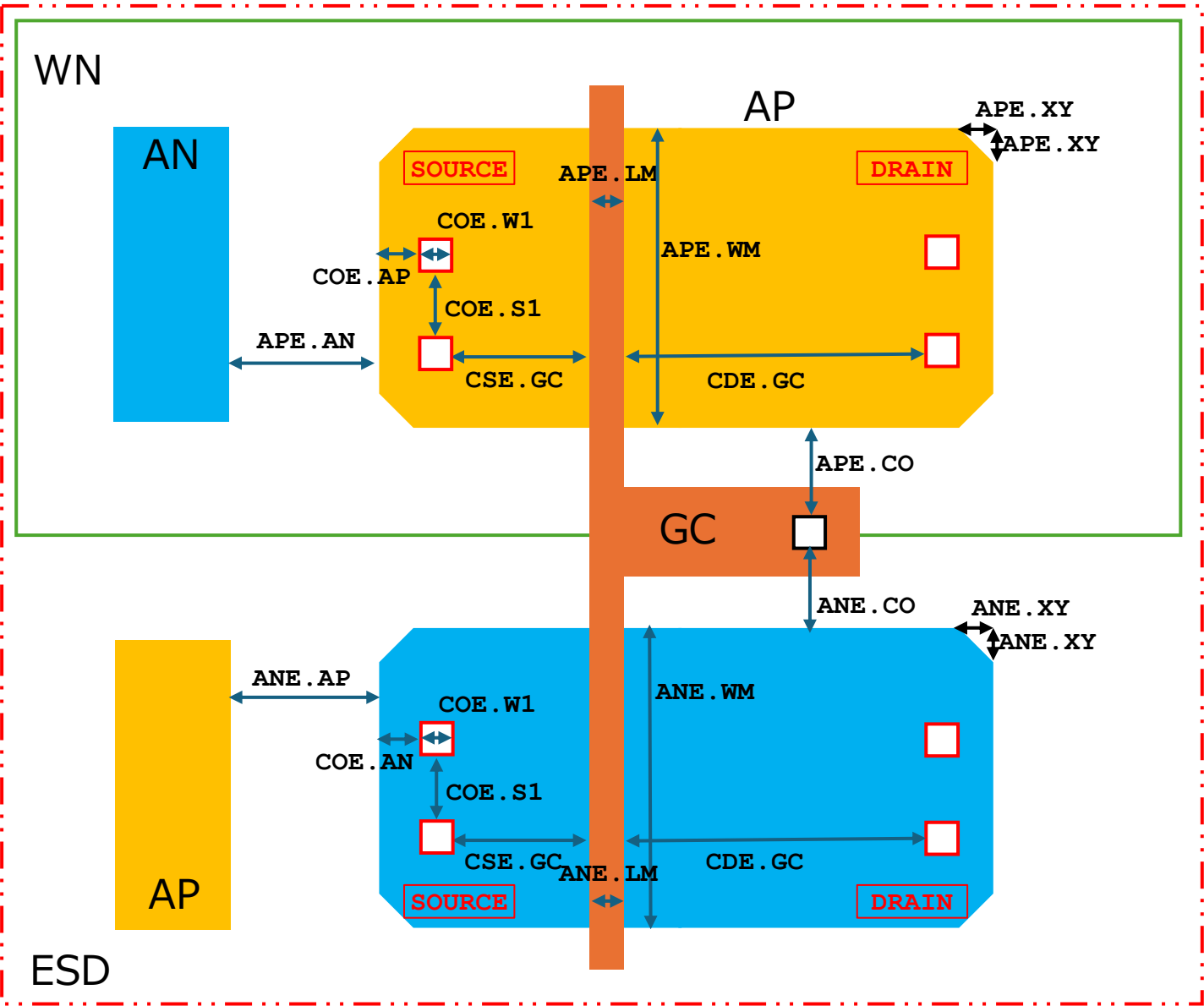


| Rule  | L1     | L2     | Func | MIN  | MAX |
|-------|--------|--------|------|------|-----|
| M1.W1 | M1     |        | Wmin | 1.8  |     |
| M1.S1 | M1     | M1     | Smin | 1.4  |     |
| M1.SC | M1 (C) | M1 (C) | Smin | 10.0 |     |
| M1.CO | M1     | CO (S) | Fmin | 0.8  |     |
| M1.CC | M1     | CO (C) | Fmin | 1.2  |     |
| M1.CL | M1     | CO (L) | Fmin | 1.2  |     |
| M1.SW | M1 (W) | M1     | Smin | 2.0  |     |
| V1.W1 | V1     |        | Wfix | 1.4  |     |
| V1.S1 | V1     | V1     | Smin | 1.5  |     |
| V1.M1 | V1     | M1     | Emin | 1.0  |     |
| V1.GC | V1     | GC     | Smin | 1.2  |     |
| V1.CO | V1     | CO     | Smin | 1.0  |     |
| V1.CL | V1     | CO (L) | Smin | 1.4  |     |
| M2.W1 | M2     |        | Wmin | 3.0  |     |
| M2.S1 | M2     | M2     | Smin | 2.0  |     |
| M2.V1 | M2     | V1     | Fmin | 1.0  |     |

M1 (C) : M1 in WN (C)  
CO (C) : CO on AC  
CO (S) : CO small contact  
CO (L) : CO large contact (only on AR)  
M1 (W) : Wide M1



# ESD MOS transistor related

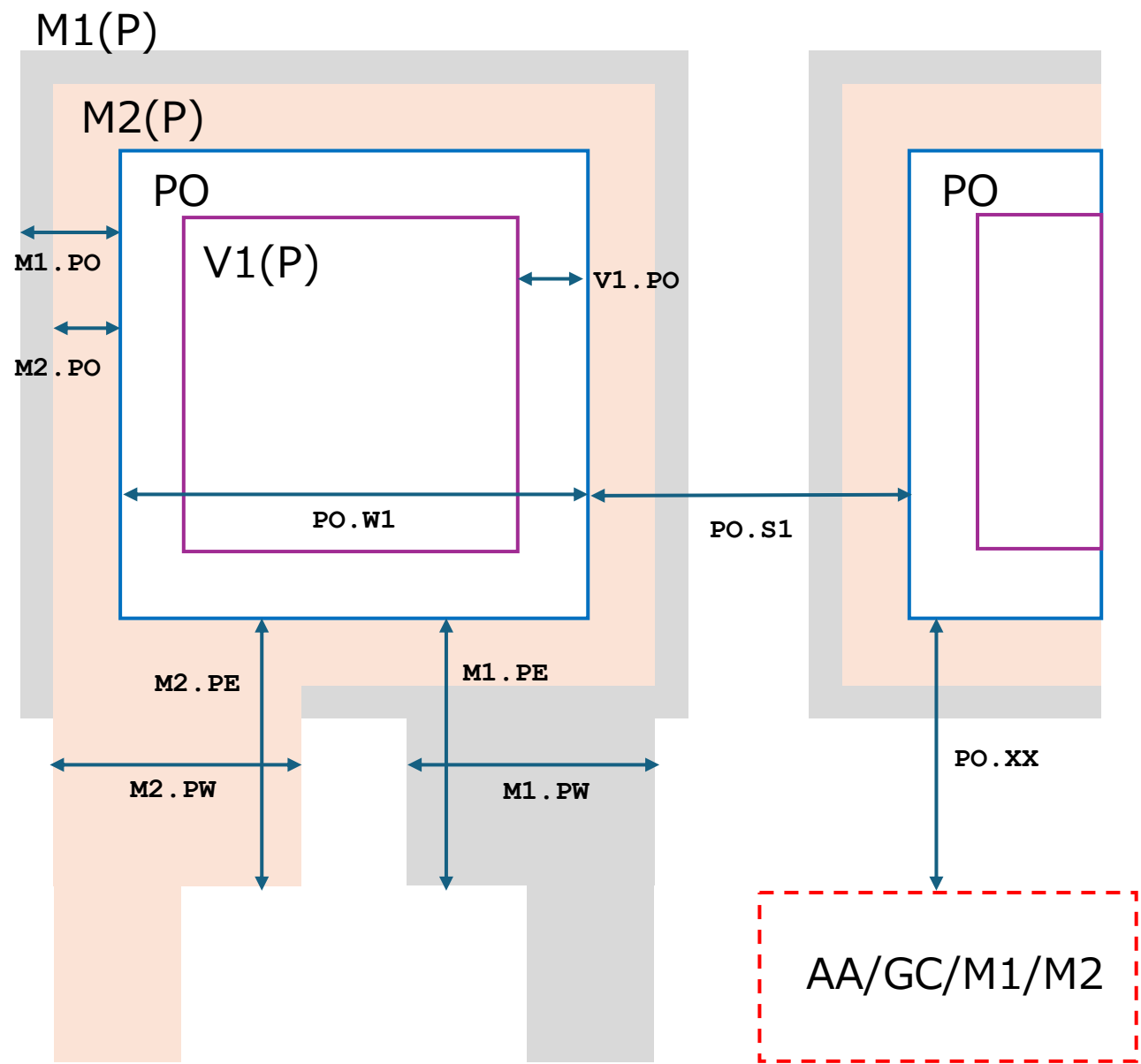


| Rule   | L1     | L2    | Func     | MIN  | MAX  |
|--------|--------|-------|----------|------|------|
| APE.WM | PMOS   | AP-GC | Wmin/max | 11.0 | 60.0 |
| APE.LM | PMOS   | GC-AN | Lmin/max | 2.0  | 30.0 |
| APE.AN | APE    | AN    | Smin     | 10.0 |      |
| APE.XY | APE    | Bevel | XYmin    | 1.4  |      |
| APE.CO | APE    | CO    | Smin     | 2.0  |      |
| ANE.WM | NMOS   | AN-GC | Wmin/max | 11.0 | 60.0 |
| ANE.LM | NMOS   | GC-AN | Lmin/max | 2.0  | 30.0 |
| ANE.AP | ANE    | AP    | Smin     | 10.0 |      |
| ANE.XY | ANE    | Bevel | XYmin    | 1.4  |      |
| ANE.CO | ANE    | CO    | Smin     | 2.0  |      |
| COE.W1 | CO (E) |       | Wfix     | 3.0  |      |
| COE.S1 | CO (E) |       | Smin     | 1.6  |      |
| CDE.GC | CD (E) | GC    | Smin     | 7.0  |      |
| CSE.GC | CS (E) | GC    | Smin     | 3.0  |      |
| COE.AP | CO (E) | AP    | Emin     | 4.0  |      |
| COE.AN | CO (E) | AN    | Emin     | 4.0  |      |

APE/ANE) : AP/AN on ESD)

CO (E) : CO on ESD

# PAD related



| Rule  | L1     | L2        | Func | MIN  | MAX |
|-------|--------|-----------|------|------|-----|
| PO.W1 | PO     |           | Wmin | 70.0 |     |
| PO.S1 | PO     |           | Smin | 64.0 |     |
| M2.PO | M2 (P) | PO        | Fmin | 5.0  |     |
| M1.PO | M1 (P) | PO        | Fmin | 5.0  |     |
| PO.V1 | PO     | V1 (P)    | Emin | 10.0 |     |
| PO.AP | PO     | AP        | Smin | 14.0 |     |
| PO.AN | PO     | AN        | Smin | 14.0 |     |
| PO.GC | PO     | GC        | Smin | 14.0 |     |
| PO.M1 | PO     | M1        | Smin | 14.0 |     |
| PO.M2 | PO     | M2        | Smin | 14.0 |     |
| M2.PE | M2     | Extention |      | 19.0 |     |
| M2.PW | M2     | Extention | Wmin | 40.0 |     |
| M1.PE | M2     | Extention |      | 19.0 |     |
| M1.PW | M2     | Extention | Wmin | 40.0 |     |

\*\* Not yet implemented

M1 (P) : M1 on PO  
M2 (P) : M2 on PO  
V1 (P) : V1 on PO

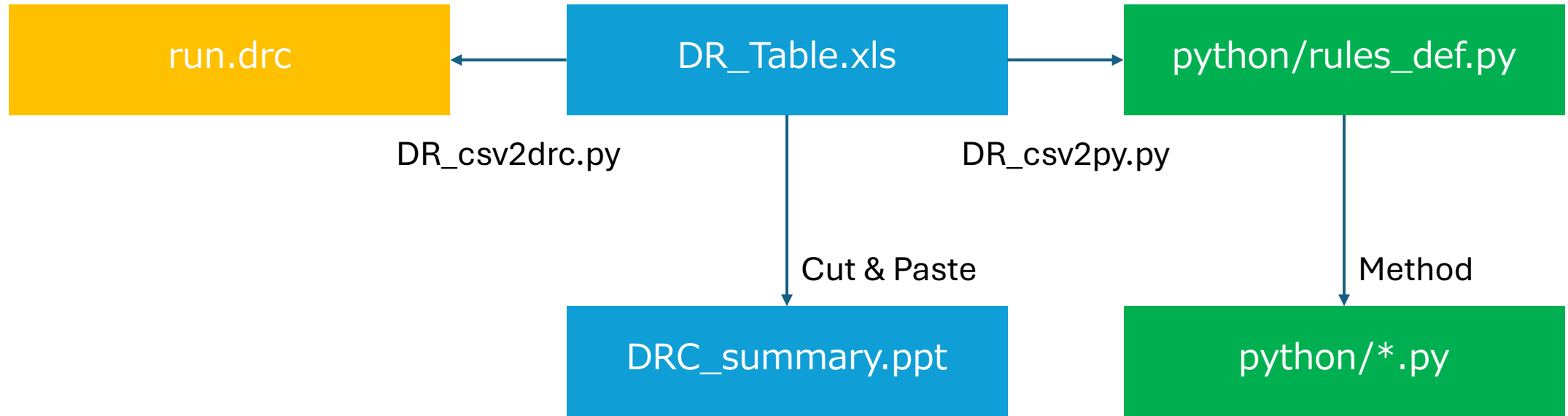
# Design Rule control

DRC integrity

# DRC

# Document

# PCell



# ALL Critical Dimension parameters are referenced by 'XX.YY' rule's name.

# Design to Tapeout Process

Mask Data Preparation

# Designer

local



[push]

remote



[action]  
Precheck  
DRC  
MDP



[clone]



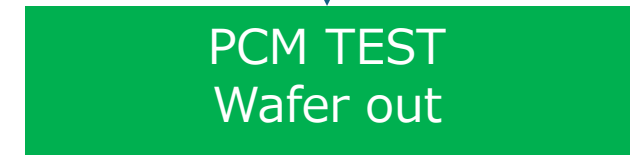
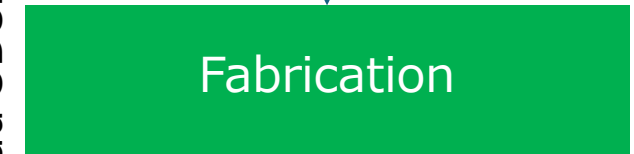
[pull request]



[action]  
Logo insertion  
Aggregation  
DRC



# Tokai Rika



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